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Grab handles for vehicles

Abstract

A steering system for a vehicle includes a steering column and a handlebar riser. The handlebar riser is coupled to the top end of the steering column and defines an internal channel. The steering system also includes a grab handle including a closed loop and a base coupled to the bottom end of the closed loop. The base is insertable into the internal channel of the handlebar riser. The steering system also includes handlebars coupled to the top end of the handlebar riser over the base of the grab handle, thereby securing the grab handle to the handlebar riser.

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Background/Summary

TECHNICAL FIELD OF THE DISCLOSURE

(1) The present disclosure relates, in general, to steering systems for vehicles and, in particular, to grab handles mounted adjacent to the handlebars of a vehicle, the grab handle including a closed loop onto which an operator of the vehicle may grasp for added stability during various vehicle maneuvers.

BACKGROUND

(2) Snowmobiles are popular land vehicles used for transportation and recreation in cold and snowy conditions. Certain snowmobiles are designed for specific applications such as trail, utility, mountain, race and crossover applications, to name a few. Snowmobiles typically include a frame assembly, or chassis, that supports various components of the snowmobile such as an engine, a transmission, a steering system and a ground-engaging endless drive track disposed in a longitudinally extending tunnel. The engine and transmission power the drive track to enable ground propulsion for the vehicle. A rider controls the operation of the snowmobile using the steering system including a handlebar assembly that is operatively linked to a pair of ski assemblies that provides flotation for the front end of the snowmobile over the snow.

(3) To provide additional stability for the rider, snowmobiles as well as other land vehicles may include a grab handle, also known as a grab bar, mountain strap or mountain handle. Grab handles may be mounted on or near the handlebars so that the rider may grasp the grab handle with one hand for stability while continuing to grasp the handlebars with the other hand for steering control. Alternatively, grab handles may also be found on the rear of the vehicle, near the seat or on the side of the vehicle for easy access. In one example application, grab handles are commonly installed on snowmobiles for use in deep snow or mountainous terrain to provide a sturdy and secure handle for riders to hold onto while standing on the vehicle and navigating steep inclines or performing maneuvers such as sidehilling, thereby keeping the rider vertical when the snowmobile is tilted. Some vehicles have a pre-installed grab handle, while for other vehicles a grab handle is installed after initial vehicle assembly. Current grab handles are either fastened directly to the handlebars or require additional parts, increasing the complexity, cost and time required to remove or install the grab handle. Relying solely on a direct connection between the grab handle and the handlebars may also lessen the stability of the grab handle, especially when loaded with the weight of the rider. Accordingly, a need has arisen for grab handles that are conveniently removable from and installable on a vehicle while providing enhanced stability to withstand high loads when utilized in challenging terrain.

SUMMARY

(4) In a first aspect, the present disclosure is directed to a steering system for a vehicle including a steering column and a handlebar riser. The handlebar riser is coupled to the top end of the steering column and defines an internal channel. The steering system also includes a grab handle including a closed loop and a base coupled to the bottom end of the closed loop. The base is insertable into the internal channel of the handlebar riser. The steering system also includes handlebars coupled to the top end of the handlebar riser over the base of the grab handle, thereby securing the grab handle to the handlebar riser.

(5) In some embodiments, the grab handle may be formed from an elastomer such as a thermoplastic elastomer. In certain embodiments, the closed loop of the grab handle may protrude from the front side of the handlebar riser. In some embodiments, the closed loop of the grab handle may have a tilted configuration such that the upper section of the closed loop is aft of the lower section of the closed loop. In certain embodiments, the closed loop of the grab handle may have a lower section having a curved profile extending forward from the base of the grab handle and curving aftward toward an upper section of the closed loop. In some embodiments, the closed loop of the grab handle may include a flat top segment. In such embodiments, the flat top segment may be substantially parallel to the handlebars. In certain embodiments, the closed loop of the grab handle may have a lower section forward of the handlebars. In some embodiments, the closed loop of the grab handle may have a lower section including a support webbing. In certain embodiments, the closed loop of the grab handle may be formed from a number of segments including left and right diagonal lower segments having bottom ends converging to the base of the grab handle, left and right diagonal upper segments having bottom ends coupled to top ends of the left and right diagonal lower segments, respectively, and a top segment. In such embodiments, the top ends of the diagonal upper segments may converge to the top segment. In some embodiments, the closed loop of the grab handle may be formed from a number of segments each having a polygonal cross-sectional shape.

(6) In certain embodiments, the base of the grab handle may have a top side defining a groove curved to contour the bottom side of the handlebars. In some embodiments, the base of the grab handle may define a cavity. In certain embodiments, the steering system may include one or more clamps coupled to the top end of the handlebar riser such that the handlebars are interposed between the one or more clamps and the handlebar riser. In some embodiments, each clamp may have a bottom side curved to contour the top side of the handlebars. In certain embodiments, the top end of the handlebar riser may define one or more left fastener holes and one or more right

fastener holes and the one or more clamps may include a left clamp and a right clamp each defining one or more fastener holes. In such embodiments, the steering system may include one or more left fasteners insertable through the one or more fastener holes of the left clamp and the one or more left fastener holes of the handlebar riser and one or more right fasteners insertable through the one or more fastener holes of the right clamp and the one or more right fastener holes of the handlebar riser, thereby coupling the clamps to the top end of the handlebar riser.
(7) In a second aspect, the present disclosure is directed to a snowmobile including a forward frame assembly and a steering system coupled to the forward frame assembly. The steering system includes a steering column and a handlebar riser. The handlebar riser is coupled to the top end of the steering column and defines an internal channel. The steering system also includes a grab handle including a closed loop and a base coupled to the bottom end of the closed loop. The base is insertable into the internal channel of the handlebar riser. The steering system also includes handlebars coupled to the top end of the handlebar riser over the base of the grab handle, thereby securing the grab handle to the handlebar riser.
(8) In some embodiments, the grab handle including the closed loop and the base may be a monolithic component. In certain embodiments, the grab handle may be formed from an injection molded polymer. In some embodiments, the closed loop of the grab handle may include a top segment substantially vertically aligned with the base of the grab handle and the handlebars such that the top segment, the base and the handlebars lie along a common lateral plane.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) For a more complete understanding of the features and advantages of the present disclosure, reference is now made to the detailed description along with the accompanying figures in which corresponding numerals in the different figures refer to corresponding parts and in which:
(2) FIGS. **1**A-**1**C are isometric and side views of a snowmobile having a grab handle in accordance with embodiments of the present disclosure;
(3) FIGS. **2**A-**2**C are various views of a steering system for a snowmobile having a grab handle in accordance with embodiments of the present disclosure;
(4) FIGS. **3**A-**3**D are various views of a handlebar assembly for a snowmobile steering system having a grab handle in accordance with embodiments of the present disclosure;
(5) FIG. **4** is an isometric view of a handlebar riser for a snowmobile steering system having a grab handle in accordance with embodiments of the present disclosure; and
(6) FIGS. **5**A-**5**C are various views of a grab handle in accordance with embodiments of the present disclosure.
DETAILED DESCRIPTION
(7) While the making and using of various embodiments of the present disclosure are discussed in detail below, it should be appreciated that the present disclosure provides many applicable inventive concepts, which can be embodied in a wide variety of specific contexts. The specific embodiments discussed herein are merely illustrative and do not delimit the scope of the present disclosure. In the interest of clarity, all features of an actual implementation may not be described in this specification. It will of course be appreciated that in the development of any such actual embodiment, numerous implementation-specific decisions must be made to achieve the developer's specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure.
(8) In the specification, reference may be made to the spatial relationships between various

components and to the spatial orientation of various aspects of components as the devices are depicted in the attached drawings. However, as will be recognized by those skilled in the art after a complete reading of the present disclosure, the devices, members, apparatuses, and the like described herein may be positioned in any desired orientation. Thus, the use of terms such as “above,” “below,” “upper,” “lower” or other like terms to describe a spatial relationship between various components or to describe the spatial orientation of aspects of such components should be understood to describe a relative relationship between the components or a spatial orientation of aspects of such components, respectively, as the devices described herein may be oriented in any desired direction. As used herein, the term “coupled” may include direct or indirect coupling by any means, including by mere contact or by moving and/or non-moving mechanical connections.
(9) Referring to FIGS. **1**A-**1**C in the drawings, a land vehicle depicted as a snowmobile is schematically illustrated and generally designated **10**. Structural support for snowmobile **10** is provided by a chassis **12** that includes a forward frame assembly **14** and a longitudinally extending tunnel **18**. Forward frame assembly **14** may be formed from interconnected tubular members such as round and hollow tubular members comprised of metal, metal alloy, polymeric materials, fiber reinforced polymer composites and/or combinations thereof that are coupled together by welds, bolts, pins or other suitable fastening means. A right side plate member **16***a* and a left side plate member **16***b* are coupled to and preferably welded to forward frame assembly **14** such that forward frame assembly **14** and plate members **16***a*, **16***b* form a welded frame assembly. Tunnel **18** is coupled to forward frame assembly **14** and/or plate members **16***a*, **16***b* with welds, bolts, rivets or other suitable means. In the illustrated embodiment, tunnel **18** includes a right sidewall **18***a*, a left sidewall **18***b* and a top panel **18***c*. Tunnel **18** may be integrally formed or may consist of multiple members that are coupled together with welds, bolts, rivets or other suitable means. Plate members **16***a*, **16***b* and tunnel **18** may be formed from sheet metal, metal alloy, fiber reinforced polymer or other suitable material or combination of materials.
(10) Various components of snowmobile **10** are assembled on or around forward frame assembly **14**. One or more body panels **20** cover and protect the various components of snowmobile **10** including parts of forward frame assembly **14**. For example, a hood panel **20***a*, a nose panel **20***b*, an upper left side panel **20***c* and a lower left side panel **20***d* shield underlying componentry from snow and terrain. Similarly, an upper right side panel and a lower right side panel (not visible) also shield underlying componentry from snow and terrain. In the illustrated embodiment, snowmobile **10** has a windshield **22** that shields the rider of snowmobile **10** from snow, terrain and frigid air during operation. Even through snowmobile **10** has been described and depicted as including specific body panels **20**, it should be understood by those having ordinary skill in the art that a snowmobile of the present disclosure may include any number of body panels in any configuration to provide the shielding functionality. In addition, it should be understood by those having ordinary skill in the art that the right side and the left side of snowmobile **10** will be with reference to a seated rider of snowmobile **10** with the right side of snowmobile **10** corresponding to the right side of the rider and the left side of snowmobile **10** corresponding to the left side of the rider.
(11) Body panels **20** as well as other components have been removed from snowmobile **10** in FIGS. **1**B-**1**C to reveal the underlying components of snowmobile **10**. For example, snowmobile **10** has a powertrain **24** that includes an engine **26** and a drivetrain **28**, both of which are coupled to forward frame assembly **14**. Engine **26** resides in a bay formed within forward frame assembly **14** of chassis **12**. Engine **26** may be any type of engine such as a four-stroke engine, a two-stroke engine, an electric motor or other prime mover. In the illustrated embodiment, engine **26** is an internal combustion engine such as a naturally aspirated internal combustion engine or a forced induction internal combustion engine that includes, for example, one or more turbochargers and/or superchargers. Drivetrain **28** includes a transmission depicted as a continuously variable transmission **30** that varies the ratio of the engine output speed to the drive track input speed. In other embodiments, the transmission for snowmobile **10** may be an electrically variable

transmission or other suitable transmission type. A drive track system **32** is at least partially disposed within and/or below tunnel **18** and is in contact with the ground to provide ground propulsion for snowmobile **10**. Torque and rotational energy are provided to drive track system **32** from powertrain **24**. Drive track system **32** includes a track frame **34**, an internal suspension **36**, a plurality of idler wheels **38** such as idler wheels **38***a*, **38***b*, **38***c*, **38***d* and an endless track **40**. Track frame **34** may be coupled to forward frame assembly **14** via a swing arm having a coil spring, a rigid strut, a torsion spring, an elastomeric member or any other suitable coupling configuration. Endless track **40** is driven by a track drive sprocket via a track driveshaft (not visible) that is rotated responsive to torque provided from continuously variable transmission **30**. Endless track **40** rotates around track frame **34** and idler wheels **38** to propel snowmobile **10** in either the forward direction, as indicated by arrow **42***a*, or the backward direction, as indicated by arrow **42***b*. When viewed from the right side of snowmobile **10**, endless track **40** rotates around track frame **34** and idler wheels **38** in the clockwise direction, as indicated by arrow **44***a*, to propel snowmobile **10** in forward direction **42***a*. Endless track **40** rotates around track frame **34** and idler wheels **38** in the counterclockwise direction, as indicated by arrow **44***b*, to propel snowmobile **10** in backward direction **42***b*. Forward and backward directions **42***a*, **42***b* also represent the longitudinal direction of snowmobile **10** with the lateral direction of snowmobile **10** being normal thereto and represented by the rightward direction, as indicated by arrow **46***a*, and the leftward direction, as indicated by arrow **46***b* in FIG. **1**A. The backward direction may also be referred to herein as the aftward direction.

(12) Snowmobile **10** has a steering system **48** that includes a handlebar assembly **50**, a steering column **52**, a steering arm assembly **54**, a right tie rod **56**, a left tie rod **58**, a right ski assembly **60** including a right spindle **60***a* and a right ski **60***b* and a left ski assembly **62** including a left spindle **62***a* and a left ski **62***b*. Right ski assembly **60** and left ski assembly **62** may be referred to collectively as the ski system of snowmobile **10**. Snowmobile **10** has a front suspension assembly **64** that is coupled between forward frame assembly **14** and ski assemblies **60**, **62** to provide front end support for snowmobile **10**. In addition, right ski assembly **60** is coupled to forward frame assembly **14** by upper and lower A-arms **66***a*, **66***b*, and left ski assembly **62** is coupled to forward frame assembly **14** by upper and lower A-arms **68***a*, **68***b*. Steering system **48** enables the rider to steer snowmobile **10** by rotating handlebar assembly **50**, which causes ski assemblies **60**, **62** to pivot. In the illustrated embodiment, the pivoting of ski assemblies **60**, **62** responsive to rotation of handlebar assembly **50** is assisted by an electric power steering system (EPS) depicted as electronic steering assist unit **70**. Handlebar assembly **50** includes handlebars controls **72** that allow the rider to control various functions of snowmobile **10** such as braking and lighting functions.

(13) The rider controls snowmobile **10** from a seat **74** that is positioned atop a fuel tank **76**, above tunnel **18**, aft of handlebar assembly **50** and aft of forward frame assembly **14**. Snowmobile **10** has a lift bumper **78** that is coupled to an aft end of tunnel **18** that enables a person to lift the rear end of snowmobile **10** in the event snowmobile **10** becomes stuck or needs to be repositioned when it is not moving. Snowmobile **10** has a snow flap **80** that deflects snow emitted by endless track **40**. A taillight housing **82** is also coupled to lift bumper **78** and houses a taillight of snowmobile **10**. Snowmobile **10** has an exhaust system **84** that includes an exhaust manifold **86** that is coupled to one or more exhaust outlets on engine **26**, an exhaust duct **88** and a muffler **90**. As exhaust system **84** including exhaust manifold **86** is coupled to the forward side of engine **26**, the forward side of engine **26** may be referred to as the hot side of engine **26** due to the hot temperatures associated with engine exhaust. The aftward side of engine **26** is concomitantly considered the cool side of engine **26** as hot exhaust system components are located opposite and/or remote therefrom.

(14) To provide additional stability for the rider, snowmobile **10** as well as other land vehicles may include a grab handle. Grab handles may be mounted on or near the handlebars so that the rider may grasp the grab handle with one hand for stability while continuing to grasp the handlebars with the other hand for steering control. For example, grab handles are commonly installed on

snowmobiles for use in deep snow or mountainous terrain to provide a sturdy and secure handle for riders to hold onto while standing on the vehicle and navigating steep inclines or performing maneuvers such as sidehilling, thereby keeping the rider vertical when the snowmobile is tilted. Some vehicles have a pre-installed grab handle, while for other vehicles a grab handle is installed after initial vehicle assembly. Current grab handles are either fastened directly to the handlebars or require additional parts, increasing the complexity, cost and time required to remove or install the grab handle. Relying solely on a direct connection between the grab handle and the handlebars may also lessen the stability of the grab handle, especially when loaded with the weight of the rider. To address these and other issues with current grab handles, steering system **48** of snowmobile **10** includes a grab handle **92** as described in the illustrative embodiments herein.

(15) It should be appreciated that snowmobile **10** is merely illustrative of a variety of vehicles that can implement the embodiments disclosed herein. Indeed, grab handle **92** may be implemented on any ground-based vehicle. Other vehicle implementations can include motorcycles, snow bikes, all-terrain vehicles (ATVs), utility vehicles, recreational vehicles, scooters, automobiles, mopeds, straddle-type vehicles, jet skis and the like. As such, those skilled in the art will recognize that grab handle **92** can be integrated into a variety of vehicle configurations. It should be appreciated that even though ground-based vehicles are particularly well-suited to implement the embodiments of the present disclosure, airborne vehicles and devices such as aircraft can also implement the embodiments.

(16) Referring additionally to FIGS. **2**A-**2**C in the drawings, further details relating to steering system **48** of snowmobile **10** will now be disclosed. As discussed herein, steering system **48** includes handlebar assembly **50**, steering column **52**, steering arm assembly **54**, right tie rod **56**, left tie rod **58**, right ski assembly **60** including right spindle **60***a* and right ski **60***b*, and left ski assembly **62** including left spindle **62***a* and left ski **62***b*. In addition, steering system **48** includes electronic steering assist unit **70**. In the illustrated embodiment, steering column **52** is a straight steering column formed as a non-segmented single post that is positioned forward of upper cross member **14***a* and along a centerline **100** of snowmobile **10**. In other embodiments, the steering column may be a segmented straight steering column that has upper and lower posts, a bent steering column including, for example, a universal joint between upper and lower posts, an articulated steering column that has multiple posts routed around other snowmobile components using multiple joints, a laterally offset steering column that extends downwardly, forwardly and laterally from handlebar assembly **50** to the lower steering assembly or other suitable connection between handlebar assembly **50** and the lower steering assembly. As best seen in FIG. **2**C, steering column **52** has a top end **52***a* and a bottom end **52***b*. Top end **52***a* of steering column **52** is coupled to handlebar riser **102** of handlebar assembly **50**. Bottom end **52***b* of steering column **52** includes a splined coupler **52***c* that may be integral with or coupled to bottom end **52***b* of steering column **52**. Splined coupler **52***c* receives an input shaft **104** having input splines **106** thereon to couple lower end **52***b* of steering column **52** to electronic steering assist unit **70**. Steering arm assembly **54** includes a splined coupler **54***a* that receives an output shaft **108** having output splines **110** thereon such that electronic steering assist unit **70** is coupled directly to steering arm assembly **54** without a steering column post or other extension positioned therebetween. In other embodiments, a steering column post or other extension may be positioned between electronic steering assist unit **70** and steering arm assembly **54**. Steering arm assembly **54** is coupled to the proximal ends of tie rods **56**, **58**. The distal ends of tie rods **56**, **58** are respectively coupled to ski assemblies **60**, **62** such that rotation of handlebar assembly **50** by the rider of snowmobile **10**, together with the assistance of electronic steering assist unit **70**, causes ski assemblies **60**, **62** to pivot, thus turning snowmobile **10**. A lower end **54***b* of steering arm assembly **54** is received within a bearing assembly (not visible) of nose truss **14***b* such that steering arm assembly **54** is operable to rotate relative thereto.

(17) Electronic steering assist unit **70** includes an outer housing **112** that contains the working components thereof including, for example, an electric motor, a torque sensor, a controller and a

torsion bar that couples input shaft **104** to output shaft **108**. In other embodiments, an electronic steering assist unit may have an alternate shaft configuration including, for example, a single piece shaft design. Outer housing **112** is fixed against rotation relative to forward frame assembly **14** by brackets **114***a*, **114***b*. In operation, the input torque applied from handlebar assembly **50** via steering column **52** on input shaft **104** is measured by the torque sensor. Input torque data is then provided to the controller from the torque sensor. Based upon the input torque data and additional factors such as the speed of snowmobile **10**, the controller commands the electric motor to provide an output assist torque to output shaft **108** that is additive to the input torque applied to output shaft **108** from input shaft **104** via the torsion bar. The use of electronic steering assist unit **70** improves the handling of snowmobile **10**, reduces fatigue associated with driving snowmobile **10** and can allow snowmobile **10** to be driven more aggressively. In addition, coupling electronic steering assist unit **70** directly to steering arm assembly **54** has numerous advantages over prior snowmobile steering systems that have electronic steering assist units including lowering the center of gravity of snowmobile **10** by positioning the electronic steering assist unit at a lowermost location of the steering column. In addition, coupling electronic steering assist unit **70** directly to steering arm assembly **54**, together with using a straight steering column **52** and having a common axis of rotation **116** shared by handlebar assembly **50**, steering column **52**, electronic steering assist unit **70** and steering arm assembly **54** that is positioned along centerline **100** (see FIG. **2**B) of snowmobile **10**, reduces the number of parts required in steering system **48** and reduces the complexity of steering system **48**, which improves the overall reliability of snowmobile **10**. Handlebar assembly **50** includes handlebar riser **102**, grab handle **92** and handlebars **118**. Handlebars **118** are coupled to the top end of handlebar riser **102**. Grab handle **92** includes a closed loop **120** with a base **122** that slides into an internal channel defined by handlebar riser **102** when handlebars **118** are removed from handlebar riser **102**. When handlebars **118** are coupled to the top end of handlebar riser **102**, base **122** of grab handle **92** is securely locked into the internal channel of handlebar riser **102** so that closed loop **120** provides a stable point onto which a rider of snowmobile **10** may grasp when navigating challenging terrain.

(18) Referring to FIGS. **3**A-**3**D in the drawings, a handlebar assembly is schematically illustrated and generally designated **200**. Handlebar assembly **200** is an example of handlebar assembly **50** in FIGS. **1**A-**1**C and **2**A-**2**C and may be implemented on any vehicle including snowmobile **10** in FIGS. **1**A-**1**C. Handlebar assembly **200** includes handlebar riser **202**, grab handle **204** and handlebars **206**. Referring additionally to FIG. **4** in the drawings, handlebar riser **202** has a top end **202***a* and a bottom end **202***b*. Bottom end **202***b* of handlebar riser **202** is coupled to the top end of a steering column such as top end **52***a* of steering column **52** in FIG. **2**C using clamps **208***a*, **208***b*. Handlebar riser **202** defines an internal channel **210** that extends from top end **202***a* to bottom end **202***b* of handlebar riser **202** such that handlebar riser **202** forms an open conduit. In other embodiments, internal channel **210** may extend only partially through handlebar riser **202**. In the illustrated embodiment, both handlebar riser **202** and internal channel **210** have a substantially rectangular cross-sectional shape when cut along elevational plane **212**, although in other embodiments handlebar riser **202** and internal channel **210** may have any cross-sectional shape including a polygonal, circular, elliptical or irregular cross-sectional shape. Handlebar riser **202** may be manufacturing using a metallic material such as steel or aluminum, a polymer or other rigid materials. Handlebar riser **202** includes four posts **214** including a forward left post **214***a*, an aft left post **214***b*, a forward right post **214***c* and an aft right post **214***d*. The top ends of posts **214** define fastener holes **216**. More specifically, forward left post **214***a* and aft left post **214***b* define forward left fastener hole **216***a* and aft left fastener hole **216***b*, respectively, and forward right post **214***c* and aft right post **214***d* define forward right fastener hole **216***c* and aft right fastener hole **216***d*, respectively. In some embodiments, fastener holes **216** may be threaded fastener holes. Interconnecting posts **214** are front, aft, left and right sides **202***c*, **202***d*, **202***e*, **202***f* of handlebar riser **202**. The top ends of front and aft sides **202***c*, **202***d* of handlebar riser **202** define grooves

218*a*, **218***b*. The top ends of left and right sides **202***e*, **202***f* of handlebar riser **202** define grooves **218***c*, **218***d* that are curved to contour the underside of handlebars **206**, which in the illustrated embodiment has a substantially circular cross-section.

(19) Referring additionally to FIGS. **5**A-**5**C in the drawings, grab handle **204** of handlebar assembly **200** includes a closed loop **220** and a base **222** coupled to the bottom end of closed loop **220**. In the illustrated embodiment, closed loop **220** and base **222** form a monolithic, or integral, grab handle **204**, although in other embodiments closed loop **220** and base **222** may be separate components coupled to one another by any means. Closed loop **220** approximates a polygonal or hexagonal shape and is formed from a number of substantially linear segments **224** including left and right diagonal lower segments **224***a*, **224***b* having bottom ends converging to base **222**. Closed loop **220** also includes left and right diagonal upper segments **224***c*, **224***d* having bottom ends coupled to the top ends of left and right diagonal lower segments **224***a*, **224***b*, respectively. The top ends of left and right diagonal upper segments **224***c*, **224***d* converge to a flat and substantially horizontal top segment **224***e*. Linear segments **224***a*, **224***b*, **224***c*, **224***d*, **224***e* are joined by curved transitions **226** to reduce jagged edges for a more comfortable hand grip. While the illustrated embodiment shows closed loop **220** to have five segments **224**, closed loop **220** may be formed from any number of segments such as three, four, six or more segments. Furthermore, while segments **224** are shown as linear segments, in other embodiments segments **224** may be curved segments. Closed loop **220** may also approximate shapes other than a polygon such as a circular, elliptical or irregular shape. As best seen in FIGS. **3**D and **5**C with regard to top segment **224***e*, each segment **224** may have a polygonal cross-sectional shape to reduce hand slippage when grasped by a rider. In the illustrated embodiment, each segment **224** has an octagonal cross-sectional shape, although in other embodiments the cross-sectional shape of each segment **224** may have any number of sides or alternatively may have a circular, elliptical or irregular cross-sectional shape. The cross-sectional shapes of segments **224** may be uniform or non-uniform.

(20) Base **222** of grab handle **204** has left, right and aft sides **222***a*, **222***b*, **222***c*. The front and bottom sides of base **222** defines a cavity **228**, the depth **228***a* of which is more than half the depth **222***d* of base **222**. As best seen in FIG. **3**D, base **222** is sized and shaped to fit within internal channel **210** of handlebar riser **202**. In particular, the outer perimeter of the cross-sectional shape of base **222** when cut along elevational plane **212** approximates the cross-sectional shape of internal channel **210** when cut along elevational plane **212**. In some embodiments, base **222** may be sized to create a friction fit between base **222** and internal channel **210**, although in other embodiments a looser fit between base **222** and internal channel **210** may be suitable. Grab handle **204** also includes a support webbing **230** interposed between left and right diagonal lower segments **224***a*, **224***b* above base **222**. The top side of support webbing **230** is V-shaped so as to not interfere with the grip of a rider. Support webbing **230** is a rigid member that enhances the structural strength of closed loop **220** to prevent breakage under loaded conditions. Grab handle **204** including closed loop **220** and base **222** may be manufactured using any additive, subtractive or formative manufacturing technique including, but not limited to, injection molding, extrusion, machining, 3D printing, laser cutting, stamping, welding or casting as well as others. Grab handle **204** may be formed from any rigid material including polymeric materials such as rubber, fiber reinforced polymer composites, metal alloy, metal or combinations thereof. Non-limiting examples of metals that may be used to form grab handle **204** include steel or aluminum. In some embodiments, grab handle **204** may be formed from an injection molded polymer or molded rubber. In one non-limiting example, grab handle **204** may be formed from an elastomer such as a thermoplastic elastomer. The outer surface of closed loop **220** may be textured, as in the case of textured rubber, to reduce grip slippage and enhance grip comfort of the rider.

(21) To install grab handle **204** in handlebar assembly **200**, base **222** is slid or inserted into the top of internal channel **210** of handlebar riser **202**. As best seen in FIG. **3**D, the bottom of closed loop **220** may come to rest on top of front side **202***c* of handlebar riser **202** within groove **218***a*. After

base **222** is inserted into internal channel **210** of handlebar riser **202**, handlebars **206** is placed atop base **222**. The top side of base **222** defines a groove **232** that is curved to contour or cradle the underside of handlebars **206**. With base **222** disposed beneath handlebars **206**, left and right clamps **234**, **236** are applied to secure handlebars **206** and retain base **222** within internal channel **210** of handlebar riser **202**. The undersides of clamps **234**, **236** are curved to define grooves that contour the top side of handlebars **206**. Clamps **234**, **236** also define fastener holes **238** though which fasteners **240** may be inserted. Fasteners **240** are inserted through fastener holes **238** in left clamp **234** and forward and aft left fastener holes **216***a*, **216***b* on top of handlebar riser **202** to secure left clamp **234** to the top end of handlebar riser **202**. Likewise, fasteners **240** are inserted through fastener holes **238** in right clamp **236** and forward and aft right fastener holes **216***c*, **216***d* on top of handlebar riser **202** to secure right clamp **236** to the top end of handlebar riser **202**. In some embodiments, fasteners **240** have outer threads and are screwed into fastener holes **216**, which may have inner threads. With clamps **234**, **236** thus secured, handlebars **206** is interposed between clamps **234**, **236** and handlebar riser **202** and handlebars **206** securely retains base **222** of grab handle **204** within internal channel **210** of handlebar riser **202** without the need to directly fasten grab handle **204** to handlebar riser **202**, handlebars **206** or other components of handlebar assembly **200** with fasteners. In contrast to some current grab handles, grab handle **204** also need not be directly molded onto adjacent components such as handlebar riser **202** or handlebars **206**. In addition, clamps **234**, **236** are used to secure handlebars **206** to handlebar riser **202** regardless of whether grab handle **204** is present on the vehicle. Because the same clamps **234**, **236** are used with or without grab handle **204**, the cost and number of parts required to install grab handle **204** on the vehicle is reduced and grab handle **204** is more easily installed or removed from the vehicle.

(22) Once handlebar assembly **200** including handlebar riser **202**, grab handle **204** and handlebars **206** is fully assembled, top segment **224***e* of closed loop **220** is substantially parallel to handlebars **206**. As best seen in FIG. **3**D, once installed, closed loop **220** has a tilted configuration such that upper section **220***a* of closed loop **220** is aft of lower section **220***b* of closed loop **220**. Lower section **220***b* of closed loop **220** extends forward from the top of base **222** and protrudes from front side **202***c* of handlebar riser **202** via groove **218***a*. Lower section **220***b* of closed loop **220** also has a curved profile that is forward of handlebars **206** and curves aftward toward upper section **220***a* of closed loop **220** so that closed loop **220** curves back toward the rider of the vehicle for easier access. Top segment **224***e* of closed loop **220** is substantially vertically aligned with handlebars **206** and base **222** of grab handle **204** such that top segment **224***e*, handlebars **206** and base **222** lie along a common lateral plane **242**. When base **222** resides in internal channel **210** of handlebar riser **202**, cavity **228** provides clearance for other components that may also be disposed in handlebar riser **202** such as the top end of the steering column.

(23) The foregoing description of embodiments of the disclosure has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise form disclosed, and modifications and variations are possible in light of the above teachings or may be acquired from practice of the disclosure. The embodiments were chosen and described in order to explain the principals of the disclosure and its practical application to enable one skilled in the art to utilize the disclosure in various embodiments and with various modifications as are suited to the particular use contemplated. For example, numerous combinations of the features disclosed herein will be apparent to persons skilled in the art including the combining of features described in different and diverse embodiments, implementations, contexts, applications and/or figures. Other substitutions, modifications, changes and omissions may be made in the design, operating conditions and arrangement of the embodiments without departing from the scope of the present disclosure. Such modifications and combinations of the illustrative embodiments as well as other embodiments will be apparent to persons skilled in the art upon reference to the description. It is, therefore, intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A steering system for a vehicle, the steering system comprising: a steering column having a top end; a handlebar riser having a top end, the handlebar riser coupled to the top end of the steering column and defining an internal channel; a grab handle including a closed loop having a bottom end and a base coupled to the bottom end of the closed loop, the base insertable into the internal channel of the handlebar riser; and handlebars coupled to the top end of the handlebar riser over the base of the grab handle, thereby securing the grab handle to the handlebar riser.
2. The steering system as recited in claim 1 wherein, the grab handle is formed from a thermoplastic elastomer.
3. The steering system as recited in claim 1 wherein, the closed loop of the grab handle protrudes from a front side of the handlebar riser.
4. The steering system as recited in claim 1 wherein, the closed loop of the grab handle has a tilted configuration such that an upper section of the closed loop is aft of a lower section of the closed loop.
5. The steering system as recited in claim 1 wherein, the closed loop of the grab handle has a lower section having a curved profile extending forward from the base of the grab handle and curving aftward toward an upper section of the closed loop.
6. The steering system as recited in claim 1 wherein, the closed loop of the grab handle comprises a flat top segment.
7. The steering system as recited in claim 6 wherein, the flat top segment is substantially parallel to the handlebars.
8. The steering system as recited in claim 1 wherein, the closed loop of the grab handle has a lower section forward of the handlebars.
9. The steering system as recited in claim 1 wherein, the closed loop of the grab handle has a lower section including a support webbing.
10. The steering system as recited in claim 1 wherein, the closed loop of the grab handle is formed from a plurality of segments, the plurality of segments comprising: left and right diagonal lower segments having bottom ends converging to the base of the grab handle; left and right diagonal upper segments having bottom ends coupled to top ends of the left and right diagonal lower segments, respectively; and a top segment; wherein, top ends of the diagonal upper segments converge to the top segment.
11. The steering system as recited in claim 1 wherein, the closed loop of the grab handle is formed from a plurality of segments each having a polygonal cross-sectional shape.
12. The steering system as recited in claim 1 wherein, the base of the grab handle has a top side defining a groove curved to contour a bottom side of the handlebars.
13. The steering system as recited in claim 1 wherein, the base of the grab handle defines a cavity.
14. The steering system as recited in claim 1 further comprising one or more clamps coupled to the top end of the handlebar riser such that the handlebars are interposed between the one or more clamps and the handlebar riser.
15. The steering system as recited in claim 14 wherein, each clamp has a bottom side curved to contour a top side of the handlebars.
16. The steering system as recited in claim 14 wherein, the top end of the handlebar riser defines one or more left fastener holes and one or more right fastener holes; wherein, the one or more clamps comprises a left clamp and a right clamp each defining one or more fastener holes; and wherein, the steering system further comprises one or more left fasteners insertable through the one or more fastener holes of the left clamp and the one or more left fastener holes of the handlebar riser and one or more right fasteners insertable through the one or more fastener holes of the right clamp and the one or more right fastener holes of the handlebar riser, thereby coupling the clamps

to the top end of the handlebar riser.
17. A snowmobile comprising: a forward frame assembly; a steering system coupled to the forward frame assembly, the steering system comprising: a steering column having a top end; a handlebar riser having a top end, the handlebar riser coupled to the top end of the steering column and defining an internal channel; a grab handle including a closed loop having a bottom end and a base coupled to the bottom end of the closed loop, the base insertable into the internal channel of the handlebar riser; and handlebars coupled to the top end of the handlebar riser over the base of the grab handle, thereby securing the grab handle to the handlebar riser.
18. The snowmobile as recited in claim 17 wherein, the grab handle including the closed loop and the base is a monolithic component.
19. The snowmobile as recited in claim 17 wherein, the grab handle is formed from an injection molded polymer.
20. The snowmobile as recited in claim 17 wherein, the closed loop of the grab handle comprises a top segment substantially vertically aligned with the base of the grab handle and the handlebars such that the top segment, the base and the handlebars lie along a common lateral plane.